

## Mathematical Reasoning Trace

Step1: Given  $x + y = 10$

DEFINITION

Step2: Given  $2x - y = 5$

DEFINITION

Step3: Adding Eq.  $3x = 15$

COMPUTATION

Step4: Therefore  $x = 5$

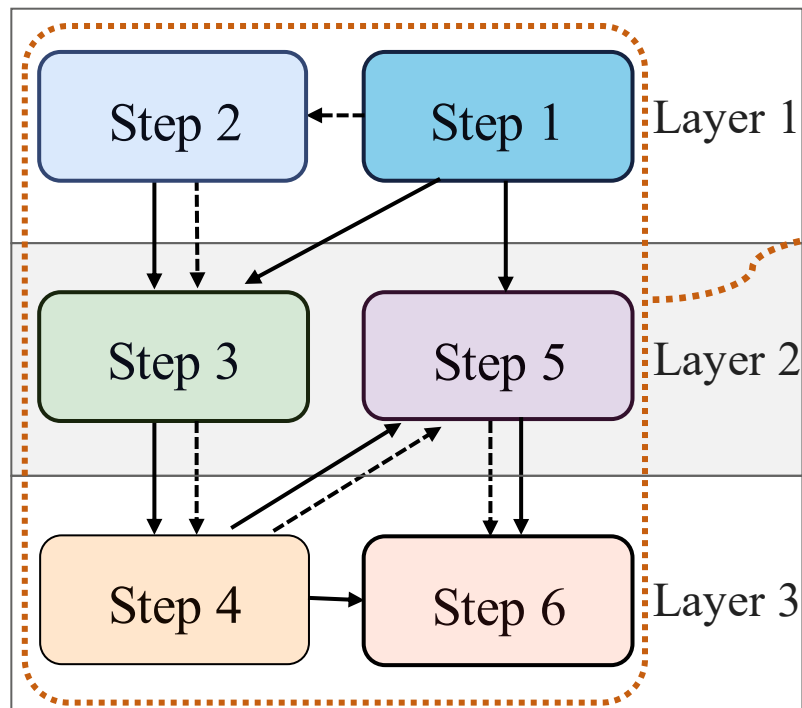
DERIVATION

Step5: Substitute  $y = 10 - 5 = 5$

SUBSTITUTION

Step6: Conclude  $x = 5, y = 5$

CONCLUSION



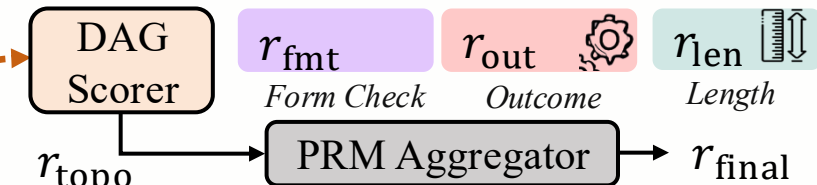
→ dependency    - - - - - sequential

[Acyclic] / Directionally Consistent

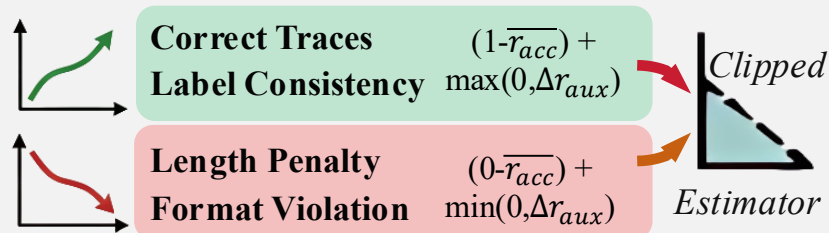
[Connected] / No Orphan Conclusion

## Topology-Guided Self-Distillation

### Implicit Process Reward Model



### Asymmetric Clipping Advantage Estimator



### Topology-Guided On-Policy Distillation

